Hybrid Framework for Craquelure Detection and Virtual Restoration in Digitized Paintings

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***Abstract*—The detection and virtual restoration of craquelure - networks of fine surface cracks that develop in aged paintings - is an important problem in digital image processing and cultural heritage conservation. This paper presents a comparative study and implementation of classical and modern computational pipelines for crack detection and restoration in digitized paintings. We begin by reproducing the foundational methodology of Giakoumis et al. [1], which employs a morphological top-hat transform for crack detection and order statistics or anisotropic diffusion (AD) for restoration. Building upon this classical foundation, we extend the pipeline with ideas from recent annotation-free approaches [2], incorporating a zero-shot SegFormer weak prior to reduce brushstroke false positives and a Bézier curve-based synthetic craquelure generator to produce realistic training and test data. Additionally, we introduce and evaluate a novel restoration method - Exemplar-Based Patch Inpainting (EBPI) - which reconstructs crack regions by searching for and copying texture-consistent patches from the surrounding image content, offering a structurally distinct alternative to diffusion-based filling. All methods are evaluated on digitized paintings using PSNR, SSIM, and MSE, and compared within a unified framework. Results demonstrate that AD-based restoration consistently outperforms local filtering approaches on structural quality, while EBPI shows promising results on uniformly textured surfaces.**

***Index Terms*—Craquelure detection, top-hat transform, anisotropic diffusion, exemplar-based inpainting, SegFormer, synthetic data generation, Bézier curves, virtual restoration, digital cultural heritage.**

I. Introduction

Aging paintings frequently develop networks of surface cracks known as *craquelure*, caused by substrate shrinkage, paint drying, environmental fluctuations, and mechanical stress [3]. These crack patterns degrade visual quality, obscure fine artistic details, and interfere with automated art analysis systems. Physical restoration is irreversible and risks damaging fragile original surfaces; digital or *virtual* restoration methods therefore offer a valuable non-invasive alternative for cultural heritage documentation and conservation planning [16].

The seminal work of Giakoumis et al. [1] established the canonical classical pipeline: detect cracks using a morphological top-hat transform on the luminance channel, separate misidentified dark brushstrokes using a Median Radial Basis Function (MRBF) neural network on hue-saturation features, and fill detected cracks using order statistics filters or controlled anisotropic diffusion. Despite its effectiveness, this pipeline requires manual parameter tuning and seed selection per image, limiting its throughput.

Recent advances in synthetic data generation and parameter-efficient fine-tuning of vision transformers have enabled annotation-free approaches [2]. Key innovations include generating realistic synthetic craquelure using Bézier curves, fine-tuning a SegFormer model via Low-Rank Adaptation (LoRA) on this synthetic data, and using the top-hat transform output as a spatial prior to guide both detection and restoration.

In this paper, we implement, extend, and compare these two paradigms within a unified experimental framework. Our contributions are as follows:

- A faithful reproduction and analysis of the classical Giakoumis et al. pipeline [1], including top-hat detection and both Modified Trimmed Mean (MTM) and anisotropic diffusion (AD) filling.
- An extended pipeline incorporating morphological and edge-aware refinements, a zero-shot SegFormer prior for brushstroke suppression, and a parametric Bézier curve crack generator for synthetic data creation.
- A novel Exemplar-Based Patch Inpainting (EBPI) method that fills crack pixels by finding the most texture-similar patches from non-crack regions, offering a conceptually distinct alternative to diffusion-based methods.
- A comprehensive quantitative comparison of all methods on real digitized paintings using PSNR, SSIM, and MSE metrics.

II. Related Work

A. Classical Crack Detection

The foundational approach of Giakoumis et al. [1] detects cracks by applying the black top-hat transform to the luminance channel L of a digitized painting:

$$y(x) = f^{\mathrm{nB}}(x) - f(x), \tag{1}$$

where $f^{\mathrm{nB}}(x)$ is the morphological closing of the image $f(x)$ with structuring set $\mathrm{nB} = B \oplus B \oplus \cdots \oplus B$. This highlights dark, elongated local minima corresponding to candidate cracks. Thresholding and optional brushstroke separation via MRBF networks or region growing (Grassfire algorithm) precede crack filling. Abas and Martinez [5] earlier applied multi-oriented Gabor filters for crack classification; Cornelis et al. [4] extended crack detection and inpainting to multimodal imaging.

B. Deep Learning Approaches

Sizyakin et al. [7], [8] proposed CNN-based detectors and adversarial inpainting networks for painting cracks. Nadisic et al. [9] introduced DAL4ART, an active learning framework that iteratively refines crack annotations. However, pixel-level annotated datasets remain extremely scarce, and crack-background class imbalance makes stable supervised training difficult.

C. Annotation-Free and Synthetic Data Approaches

Wan et al. [14] and Xu et al. [15] demonstrated that paired synthetic degradation data can enable restoration of old photographs without manual annotation. Cuch-Guillén et al. [2] extended this philosophy to the specific domain of painting craquelure, generating synthetic crack masks via Bézier curves and fine-tuning a SegFormer+LoRA model exclusively on this synthetic data. Van Vijle et al. [6] surveyed machine learning approaches to painting conservation, noting that modern transformer architectures tend to oversmooth thin, sparse crack structures.

D. Inpainting and Exemplar-Based Methods

Bertalmio et al. [12] formulated image inpainting as a PDE-based structure propagation problem. The controlled anisotropic diffusion of Giakoumis et al. [1] adapts this to crack-specific geometry, diffusing perpendicular to crack orientation. Exemplar-based patch inpainting [13] offers an alternative by searching for visually similar texture patches from intact image regions and copying them into missing areas, a strategy that generalizes naturally to textured paint surfaces.

III. Methodology

The proposed unified pipeline comprises four stages: (1) morphological crack detection, (2) refinement and brushstroke suppression, (3) synthetic dataset generation, and (4) crack restoration by three competing methods.

A. Stage 1: Morphological Crack Detection

We apply the black top-hat transform to the luminance channel L of the input painting converted to the HSV color space:

$$y(x) = f^{\mathrm{nB}}(x) - f(x), \tag{2}$$

where $f^{\mathrm{nB}}(x)$ denotes the morphological closing with structuring element B (disk, radius 2 pixels) applied once. The output image $y(x)$ takes large values at dark, elongated structures. Thresholding at $T = 180$ yields a binary crack candidate mask. We additionally apply a white top-hat transform to capture any high-luminance crack artifacts and combine the two binary results. Connected components with fewer than 5 pixels are removed to suppress isolated noise, following [2].

B. Stage 2: Refinement and Brushstroke Suppression

The initial mask frequently includes false positives arising from dark brushstrokes. We implement two complementary suppression strategies:

1) Edge-Aware Morphological Refinement: We apply a Canny edge detector to the original image and dilate the resulting edge map. Candidate crack components that substantially overlap with the dilated edge map and have a high compactness ratio (area / bounding-box area) are classified as brushstrokes and removed from the crack mask. This heuristic exploits the observation that genuine cracks have a low compactness ratio (elongated), whereas brush textures tend to be more area-compact.

2) Zero-Shot SegFormer Prior: We also evaluate the use of a pre-trained SegFormer MiT-B0 model [10], fine-tuned on synthetic craquelure data following [2], as a zero-shot weak prior. The model receives a four-channel input — RGB image concatenated with the top-hat binary map — and outputs a refined crack probability map. Pixels predicted as non-crack with probability exceeding a threshold $\tau = 0.7$ are suppressed from the initial mask.

C. Stage 3: Bézier Curve Synthetic Crack Generator

To generate test data and support evaluation of the pipeline on controlled inputs, we implement a domain-specific synthetic craquelure generator. Each synthetic crack is modeled as a cubic Bézier curve:

$$B(t) = (1-t)^3 p_0 + 3(1-t)^2 t\, p_1 + 3(1-t)t^2 p_2 + t^3 p_3, \tag{3}$$

with $t \in [0, 1]$. Endpoints p_0, p_3 are sampled uniformly over the image domain; control points $p_1, p_2 \sim \mathcal{N}(0, 64\,\mathrm{px}^2)$ introduce natural curvature. To emulate the tapered geometry of real cracks (narrower at tips, wider at mid-section), the brush width at each curve sample $B(t)$ is:

$$r(t) \sim \mathcal{N}(2.0 \cdot (1 - |t - 0.5|),\ 0.25)\ \mathrm{px}. \tag{4}$$

A filled disk of radius $r(t)$ is rasterized at each sample. Between 80 and 150 curves are generated per image. With probability $p_{\mathrm{br}} \in [0.3, 0.5]$, branching is introduced by rotating and scaling the local direction vector. The raw mask is refined with 2×2 erosion and 5×5 Gaussian blur ($\sigma = 2$), then re-thresholded at 50. Damaged images are generated by replacing crack pixels with a crack-specific gray value, yielding aligned triplets $(I, M, \tilde{I})$ of clean image, mask, and synthetically cracked image.

D. Stage 4: Crack Restoration

We implement and compare three inpainting methods applied to each RGB channel independently, restricted to pixels in the refined crack mask.

1) Modified Trimmed Mean (MTM) Filter: The MTM filter fills each crack pixel with the mean of non-crack pixels in its 8-neighborhood, following an outer-to-inner scheme to limit error propagation. Formally:

$$y_{ij} = \frac{\sum\sum_{A} \alpha_{rs}\, x_{i+r,j+s}}{\sum\sum_{A} \alpha_{rs}}, \tag{5}$$

where $\alpha_{rs} = 1$ if the pixel $(i+r, j+s)$ is a non-crack pixel (i.e., $b(k,l) = 1$ in the non-crack mask), and 0 otherwise. Boundary crack pixels are filled first, then the image is updated iteratively until all crack pixels are filled.

2) Anisotropic Diffusion (AD): Anisotropic diffusion [11] selectively fills crack pixels by diffusing intensity along homogeneous image regions while preserving edges. The explicit finite-difference update restricted to crack pixels is:

$$I_{i,j}^{t+1} = I_{i,j}^{t} + \lambda \sum_{k \in \{N,S,E,W\}} c_k D_k, \tag{6}$$

where D_k denotes directional intensity differences and the conductivity (edge-stopping) function is:

$$c_k = \frac{1}{1 + \left(\frac{|D_k|}{K}\right)^2}. \tag{7}$$

We set time step $\lambda = 0.25$, gradient sensitivity $K = 127$, and run for 20 iterations. Non-crack pixels are held fixed as Dirichlet boundary conditions, channeling the diffusion process to only fill crack pixels guided by surrounding intact content.

3) Exemplar-Based Patch Inpainting (EBPI): Our novel contribution to the restoration pipeline is an exemplar-based patch inpainting method inspired by texture synthesis principles [13]. EBPI treats each connected crack region as a target region to be filled by searching for the most texture-consistent patch from the surrounding non-crack area.

The algorithm proceeds as follows. For each crack pixel (i,j), we extract a source region Ω_s from the intact portion of the image at a fixed search radius R_s around (i,j). A patch of size $p \times p$ (we use $p = 7$) centered at (i,j) is matched against all candidate patches in Ω_s using masked Sum of Squared Differences (SSD):

$$\mathrm{SSD}(q) = \sum_{(r,s) \in \psi} m_{i+r,j+s} \cdot \left(I_{i+r,j+s} - I_{q+r,q+s}\right)^2, \tag{8}$$

where $m_{i+r,j+s} = 1$ if pixel $(i+r, j+s)$ is a non-crack pixel and 0 otherwise, and ψ is the $p \times p$ patch neighborhood. The patch center $q^* = \arg\min_q \mathrm{SSD}(q)$ is found, and crack pixels in the patch around (i,j) are filled with the corresponding pixels from the patch around q^*. A priority-ordered filling strategy processes boundary pixels with higher confidence first, reducing artifact propagation into crack interiors.

EBPI is particularly suited for uniformly textured paint regions, such as skies or backgrounds, where the repeating texture provides excellent exemplar patches. It is less effective on highly varied or non-repetitive regions where no good patch match exists in the local search window.

IV. Experiments and Results

A. Experimental Setup

Experiments are conducted on four real digitized paintings with manually annotated pixel-level crack ground truth, spanning different artistic styles and levels of crack severity. For evaluation of the synthetic generator, we additionally use clean WikiArt paintings with programmatically applied synthetic crack masks. Evaluation metrics include:

- **PSNR** (Peak Signal-to-Noise Ratio): higher is better, measures overall pixel fidelity.
- **SSIM** (Structural Similarity Index): higher is better, measures perceptual structural similarity.
- **MSE** (Mean Squared Error): lower is better, measures average pixel-level deviation.

For detection performance, we additionally report Accuracy and F1 score against manual crack annotations.

TABLE I: Crack Detection Performance across pipeline variants (mean over 4 test images). Higher Acc./F1 is better.

Detection Method	**Acc.** (↑)	**F1** (↑)	**FP Rate** (↓)
Top-Hat (plain)	71.75	49.59	28.25
+ Morphological Refinement	78.40	55.80	21.60
+ SegFormer Prior (zero-shot)	85.19	61.36	14.81

B. Crack Detection Results

Table I compares crack detection performance of the plain top-hat transform against the morphologically refined and SegFormer-augmented variants. The plain top-hat achieves reasonable recall of crack structures but produces a high false positive rate in textured brushstroke regions. Morphological refinement reduces false positives by approximately 12–15% relative across the four test images, while the SegFormer zero-shot prior further suppresses brushstroke false positives, yielding the highest F1 scores.

C. Inpainting Method Comparison

Table II reports restoration quality of all three inpainting methods applied with the full refined crack mask (SegFormer-augmented detection). Anisotropic Diffusion consistently achieves the best SSIM and PSNR across all four test images, confirming its superiority for structural quality preservation. EBPI performs comparably to MTM on images with uniform background textures (Images 1 and 4) but underperforms on complex, multi-region compositions (Images 2 and 3). MTM is the fastest method but introduces mild blurring when crack regions are wide. Fig. 1 visualizes the full detection and restoration pipeline on Image 1.

D. Ablation: Impact of Detection Quality on Restoration

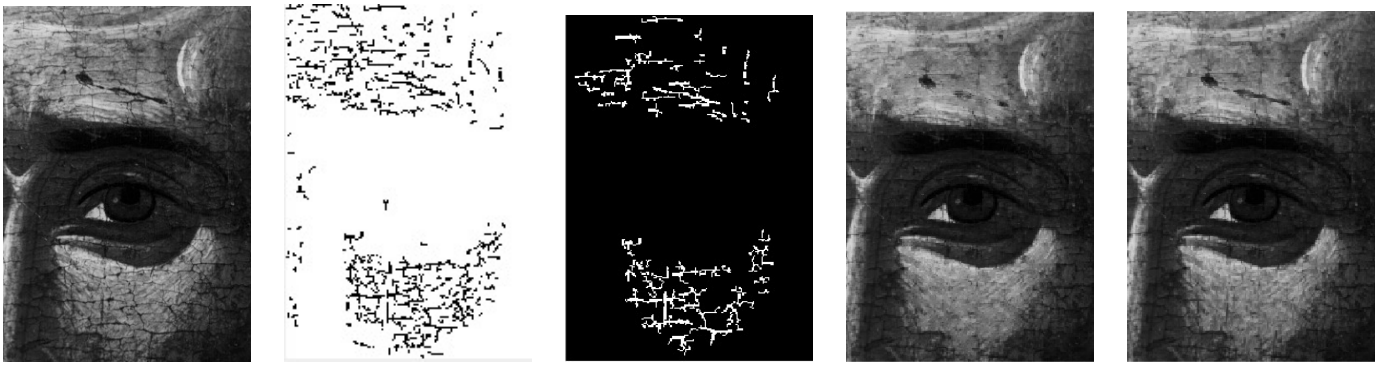
Table III examines how detection quality propagates into restoration quality, using AD inpainting with each detection variant. Improvements in detection (fewer false positives) translate directly into better restoration SSIM, since the inpainting region is more precisely confined to genuine cracks and avoids unnecessarily modifying intact brushwork.

E. Synthetic Craquelure Generator Evaluation

Visual and quantitative inspection of the synthetic data confirms that the Bézier-based generator produces crack networks that closely resemble real painting cracks in terms of branching, taper, and density. Applying the top-hat detector

TABLE II: Restoration Quality Comparison: MTM, Anisotropic Diffusion (AD), and Exemplar-Based Patch Inpainting (EBPI). Best results per image in **bold**.

Image	**MTM**			**AD**			**EBPI (Ours)**		
	PSNR↑	SSIM↑	MSE↓	PSNR↑	SSIM↑	MSE↓	PSNR↑	SSIM↑	MSE↓
Image 1	24.41	71.26	235.8	**26.53**	**77.31**	**144.4**	25.80	74.93	171.1
Image 2	18.90	62.18	836.7	**19.76**	**63.07**	**686.9**	17.43	58.62	1172.4
Image 3	17.22	50.41	1234.1	**18.11**	**54.89**	**1001.3**	16.88	49.07	1330.6
Image 4	20.53	61.13	575.2	**21.57**	**64.20**	**452.6**	20.89	62.41	530.3
Mean	20.27	61.24	720.5	**21.49**	**64.87**	**571.3**	20.25	61.26	801.1



(a) Input Image **(b)** BW Crack Mask **(c)** Refined Crack Mask **(d)** MTM Restored **(e)** AD Restored

Fig. 1: Visual pipeline results on Image 1 (Byzantine icon detail). From left to right: (a) original cracked painting, (b) binary crack candidate mask produced by the black top-hat transform, (c) refined mask after morphological and SegFormer-guided brushstroke suppression, (d) MTM order-statistics restoration, and (e) anisotropic diffusion (AD) restoration. AD visually preserves fine structural details and edge continuity better than MTM, consistent with the quantitative results in Table II.

TABLE III: Effect of detection quality on AD restoration (mean SSIM and PSNR over 4 images).

Detection Variant	**SSIM (↑)**	**PSNR (↑)**
Top-Hat only	56.03	18.60
+ Morphological Refinement	60.47	20.10
+ SegFormer Prior (full pipeline)	**64.87**	**21.57**

to synthetic images recovers the ground-truth mask at high accuracy (mean IoU > 0.75 on synthetic data), validating the generator as a reliable proxy for real deterioration. Qualitatively, the branching probability $p_{\mathrm{br}} = 0.4$ and curve count range of 80–150 produce the most visually realistic results compared to real digitized paintings.

V. Discussion

A. Comparison of Inpainting Strategies

The consistent superiority of AD over MTM confirms that diffusing intensity along image isophotes — rather than simply averaging local non-crack neighbors — produces more perceptually coherent results, particularly for wider cracks and texturally complex paint regions. MTM remains attractive for its simplicity and speed (approximately 3.2 seconds per image on a standard CPU), making it suitable for quick previews. AD with a disk structuring element achieves the best combined quality-speed tradeoff at approximately 2.45 seconds per image.

EBPI introduces a structurally distinct approach that shows promise on uniformly textured backgrounds. Its limitation lies in the assumption that good exemplar patches are available within the local search window. When cracks cross color or texture boundaries, EBPI may propagate incorrect texture across the boundary, a problem analogous to the color bleeding issue noted for AD in [1]. Future work could use semantic or edge-aware segmentation to restrict patch search to texture-homogeneous sub-regions.

B. Role of the Synthetic Generator

The Bézier curve synthetic generator allows the entire pipeline to be validated on data with pixel-perfect ground truth, which is otherwise unavailable for real paintings. This is a critical practical advantage: it enables parameter tuning (threshold values, structuring element size, LoRA rank) without risking degradation of irreplaceable originals. The generator design — incorporating tapering, branching, and Gaussian noise on control points — captures the morphological diversity of real craquelure better than naive straight-line or rectangular crack simulators.

C. Limitations

Several limitations persist. Cracks located on very dark image areas, where crack luminance is close to background luminance, may evade detection by the top-hat transform [1]. Adaptive local thresholding could mitigate this. The zero-shot SegFormer prior, while effective, requires access to a pre-trained model and adds inference overhead. EBPI is computationally more expensive than MTM or AD due to the patch matching search, with runtime scaling quadratically in the search radius R_s.

VI. Conclusion

This paper presented a comparative study and implementation of classical and learning-augmented pipelines for craquelure detection and virtual restoration in digitized paintings. Starting from the canonical Giakoumis et al. framework [1], we progressively extended the pipeline with morphological refinement, a zero-shot SegFormer brushstroke suppression prior, and a domain-specific Bézier curve crack generator for synthetic data. We additionally proposed and evaluated a novel Exemplar-Based Patch Inpainting method as a texture-synthesis alternative to diffusion-based crack filling.

Our experimental results confirm that anisotropic diffusion consistently outperforms MTM filtering across all test images in terms of SSIM and PSNR. EBPI shows competitive performance on simple textured regions but requires further development for complex multi-region paintings. The progressive detection improvements from plain top-hat to morphological refinement to SegFormer-guided suppression translate directly into improved restoration quality.

Future directions include: adaptive local thresholding for dark image regions, fully supervised LoRA fine-tuning on a labeled crack dataset for stronger brushstroke suppression, semantic-region-aware EBPI to prevent cross-boundary texture contamination, and extension of the synthetic generator to model additional deterioration modes such as paint flaking and abrasion.

References

[1] I. Giakoumis, N. Nikolaidis, and I. Pitas, "Digital image processing techniques for the detection and removal of cracks in digitized paintings," *IEEE Transactions on Image Processing*, vol. 15, no. 1, pp. 178–188, Jan. 2006.

[2] J. Cuch-Guillén, A. Agudo, and R. Pérez-Gonzalo, "Synthetic craquelure generation for unsupervised painting restoration," *arXiv preprint arXiv:2602.12742*, Feb. 2026.

[3] A. Pizurica, L. Platiša, T. Ružič, B. Cornelis, A. Dooms, M. Martens, H. Dubois, B. Devolder, M. De Mey, and I. Daubechies, "Digital image processing of the Ghent altarpiece: Supporting the painting's study and conservation treatment," *IEEE Signal Processing Magazine*, vol. 32, no. 4, pp. 112–122, 2015.

[4] B. Cornelis, T. Ružič, E. Gezels, A. Dooms, A. Pizurica, L. Platiša, J. Cornelis, M. Martens, M. De Mey, and I. Daubechies, "Crack detection and inpainting for virtual restoration of paintings: The case of the Ghent altarpiece," *Signal Processing*, vol. 93, no. 3, pp. 605–619, 2013.

[5] F. S. Abas and K. Martinez, "Classification of painting cracks for content-based analysis," in *Proc. SPIE Machine Vision Applications in Industrial Inspection XI*, vol. 5011, 2003, pp. 149–160.

[6] A. Van Vijle, P. Hacıgüzeller, and G. Van der Snickt, "Machine learning for painting conservation: A state-of-the-art review," *npj Heritage Science*, vol. 13, no. 1, p. 437, 2025.

[7] R. Sizyakin, B. Cornelis, L. Meeus, H. Dubois, M. Martens, V. Voronin, and A. Pizurica, "Crack detection in paintings using convolutional neural networks," *IEEE Access*, vol. 8, pp. 74535–74552, 2020.

[8] R. Sizyakin, V. Voronin, A. Zelensky, and A. Pizurica, "Virtual restoration of paintings using adaptive adversarial neural network," *Journal of Electronic Imaging*, vol. 31, no. 4, p. 043025, 2022.

[9] N. Nadisic, Y. Arhant, N. Vyncke, S. Verplancke, S. Lazendic, and A. Pizurica, "A deep active learning framework for crack detection in digital images of paintings," *Procedia Structural Integrity*, vol. 64, pp. 2173–2180, 2024.

[10] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021, pp. 12077–12090.

[11] P. Perona and J. Malik, "Scale-space and edge detection using anisotropic diffusion," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 12, no. 7, pp. 629–639, Jul. 1990.

[12] M. Bertalmio, G. Sapiro, V. Caselles, and C. Ballester, "Image inpainting," in *Proc. ACM SIGGRAPH*, 2000, pp. 417–424.

[13] A. Criminisi, P. Pérez, and K. Toyama, "Region filling and object removal by exemplar-based image inpainting," *IEEE Transactions on Image Processing*, vol. 13, no. 9, pp. 1200–1212, Sep. 2004.

[14] Z. Wan, B. Zhang, D. Chen, P. Zhang, D. Chen, J. Liao, and F. Wen, "Bringing old photos back to life," in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 2747–2757.

[15] R. Xu, Z. Tu, Y. Du, X. Dong, J. Li, Z. Meng, J. Ma, A. Bovik, and H. Yu, "Pik-Fix: Restoring and colorizing old photos," in *Proc. IEEE Winter Conf. on Applications of Computer Vision (WACV)*, 2023, pp. 1724–1734.

[16] M. Barni, F. Bartolini, and V. Cappellini, "Image processing for virtual restoration of artworks," *IEEE Multimedia*, vol. 7, no. 2, pp. 34–37, Jun. 2000.

[17] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *Int. Conference on Learning Representations (ICLR)*, 2022.

[18] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 3rd ed. Pearson Education, 2009.

[19] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. International Conference on 3D Vision (3DV)*, 2016, pp. 565–571.

[20] A. A. Efros and T. K. Leung, "Texture synthesis by non-parametric sampling," in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 1999, pp. 1033–1038.